Sentinel: Entity-Aware Fraud Scoring and Device-Fingerprint Ring Detection with Gated, Audited Response

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***Abstract*—Fraud-detection systems are usually reported as a single scoring model, which leaves two practical questions unexamined: whether personalizing a score to an entity's own history helps once real history depth is accounted for, and whether entities linked by a shared identifier can be called a fraud ring without verified labels. We present Sentinel, a single-author pipeline that scores transactions with an Isolation Forest, links entities into candidate rings through shared device fingerprints, and gates every automated action behind an audited, exposure-aware response policy. On the held-out split of the public IEEE-CIS benchmark (85,004 transactions, 54,963 entities), the population-only detector reaches a PR-AUC of 0.166, about three times the best rule or statistical baseline (0.055). Adding entity-deviation features does not help and slightly hurts (paired entity-bootstrap difference 0.010, 95% CI 0.008 to 0.012). We trace this to history depth: the median entity with any history has two prior transactions, and 40.9% have exactly one. Device-fingerprint linking forms 516 clusters whose pooled fraud rate is 11.6%, 2.5 times the 4.57% base rate, and a credibility filter keeps 70 of them, about twice the 34 expected under an independence null. Run over all 85,004 transactions, the response layer declines or escalates 2,259; its label-free score trigger alone flags transactions at a 27.6% fraud rate, six times the base rate, and a replay of the gate rules reproduces every logged decision, allows included, with zero discrepancies. Every ring statistic is reported as a derived proxy, not verified ground truth.**

***Index Terms*—fraud detection, anomaly detection, Isolation Forest, entity resolution, device fingerprinting, class imbalance, auditable automation**

I. Introduction

Fraud-detection systems are usually judged on one number: how well a single model separates fraudulent from legitimate transactions. Two assumptions tend to travel with that number unexamined. The first is that scoring a transaction against the entity's *own* history is always worth the engineering effort, when a large share of entities in real data simply lack enough prior transactions to baseline against. The second is that entities linked by a shared signal, such as a device fingerprint, can be reported as a fraud ring, when a shared fingerprint is a correlation that a family computer or a shared workplace terminal produces just as easily as a criminal group.

Sentinel is built around testing both assumptions rather than adopting them. It is a single-author pipeline with three subsystems: an entity- and population-aware anomaly scorer, a device-fingerprint ring detector that links entities through a bounded-prevalence identifier, and a gated response orchestrator that acts autonomously only within a defined exposure threshold, escalates everything above it to a human reviewer, and logs every decision it makes, including the decision to do nothing.

This paper makes four contributions:

1) A five-method rule- and statistics-based baseline battery, reported with full confusion matrices before any learned model is introduced, so later gains are measured against a real floor.
2) An Isolation Forest comparison of population-only against population-plus-entity-deviation features, with entity-bootstrap intervals and a history-depth diagnostic that explains the null result in the data instead of discarding it.
3) A device-fingerprint ring detector evaluated at full test scale, with a label-free enrichment measure, a chance baseline for its credibility filter, and a proxy label on every ring statistic at the point of use.
4) A gated, audited orchestrator run over all 85,004 test transactions, whose decisions are reproducible from the logged inputs, with the label-dependence of each reported rate stated explicitly.

II. Related Work

Unsupervised anomaly detection for fraud and intrusion data is well established. Isolation Forest [1] isolates anomalies through random recursive partitioning and needs no labeled attack examples, which is why it is the learned detector here. Graph-based approaches that link entities through shared attributes before scoring the resulting structure have been explored directly: Shrivastava and Pathan [2] apply a graph convolutional network to identify linked fraudulent activities, and Pourhabibi et al. [3] review graph-based anomaly detection for fraud systematically. Sentinel applies the same shared-attribute linking idea with a simpler, bounded-prevalence graph

and no learned model. Network-derived signals can also add to transaction-level scoring: Van Vlasselaer et al. [4] combine intrinsic transaction features with features from a cardholder–merchant network and report that the combination gives their best models. The per-entity history signal studied here behaves differently on this benchmark (Section IV-C). Bhattacharyya et al. [5] compare support vector machines, random forests, and logistic regression on real credit card data; Sentinel instead fixes an unsupervised detector and measures it against simple rule and statistical floors. Because fraud data are heavily imbalanced, precision-recall curves are the more informative surface than ROC when positives are rare [6], which motivates reporting PR-AUC throughout Section IV. The dataset is the public IEEE-CIS Fraud Detection benchmark [7], released by Vesta Corporation through the IEEE Computational Intelligence Society.

What is less common in this literature is pairing a learned detector with an explicit diagnostic for *why* a personalization feature does or does not help, and pairing a ring-detection layer with a point-of-use disclaimer that its linkage signal is a proxy. Sentinel treats both as reporting obligations rather than footnotes.

III. Proposed System and Methodology

A. Problem Setting and Dataset

Sentinel is evaluated on the IEEE-CIS Fraud Detection dataset [7], a public e-commerce transaction benchmark with per-transaction amount, timing, card, address, and device fields. The dataset has no account identifier, so a proxy entity is reconstructed by concatenating `card1` to `card6`, `addr1`, `addr2`, and an adjusted first-seen day (the transaction day minus `D1`), which separates accounts that share identical card and address fields. A row missing any key field receives its own unique entity, so missing data never merges unrelated records. These entities are proxies, not verified accounts.

Data are split at the entity level and in time. Entities are ranked by the timestamp of their earliest transaction and cut into contiguous blocks: the earliest 64% train the models, the next 16% calibrate the confidence threshold, and the latest 20% form the held-out test split. All transactions of an entity follow its partition, so the detector is always scored on accounts it never saw, at the accepted cost that some later transactions of an early entity fall inside a later time window. The test split holds 85,004 transactions from 54,963 entities. Entity-history features use only strictly earlier transactions of the same entity, so a test row is *has-history* when its entity has at least one earlier transaction. 30,041 rows (35.3%) qualify; the remaining 54,963 first-seen rows are cold-start. Code, configuration, and result files are public at https://github.com/devyanibishnoi/sentinel.

B. Design Principles

Four principles shape every subsystem. The detection schema is the contract between subsystems, so every score is traceable to a named feature set. No result exists without a configuration that regenerates it. Baselines come before cleverness: a learned detector is introduced only after simple rule and statistical alternatives have been run and reported in full. And every automated action is audited, including the decision to do nothing. Fig. 1 shows how the subsystems connect.

C. Baseline Battery

Before any learned model, five baselines are evaluated on the full test split: a *random prior* (flags at the training fraud rate), *rule population* (amount above the training 95th percentile), *statistical population* (more than two standard deviations from the training mean), *rule entity* (a new record-high amount for that entity), and *statistical entity* (more than two standard deviations from that entity's own prior mean). Each also yields a continuous score for PR-AUC (raw amount, absolute z-score, ratio to the entity's prior maximum, absolute entity z-score, and a uniform random draw). The entity rules need at least one (rule) or two (statistical) prior transactions, so they are evaluated on smaller populations.

D. Unsupervised Detection Layer

An Isolation Forest [1] is trained in two feature configurations. The population features are the transaction amount, hour of day, `card1`, `card2`, `card3`, `card5`, `addr1`, `addr2`, the fourteen Vesta count features `C1` to `C14`, and integer codes for `ProductCD`, `card4`, and `card6`, 25 in all, with missing values filled by training-split medians. The wider D, M, and V engineered columns are left out because of messier missingness, a named limitation. The entity-deviation configuration adds five features built from strictly earlier same-entity transactions: prior count, a has-history flag, a two-or-more-priors flag, the amount's z-score against the entity's own history, and its ratio to the entity's prior maximum, with neutral values when history is too thin. Both configurations use 200 trees, automatic contamination, and seed 42, so any PR-AUC difference reflects the feature set. The anomaly score is the negated Isolation Forest score. PR-AUC is computed on the full test split and on has-history rows only, so the entity feature is judged on the population it could help.

E. Device-Fingerprint Ring Detection

Within the test split, entities and `DeviceInfo` strings form a bipartite graph with an edge whenever an entity used a fingerprint. A fingerprint is kept only if 2 to 20 entities share it: fewer carries no linking information, and more is a generic value such as a popular handset. Connected components are kept as clusters when they hold 2 to 50 entities, because chaining through many small fingerprints can inflate a component into shared-infrastructure noise. Each cluster receives a *proxy fraud rate*, the share of its transactions carrying the dataset's fraud label, flagged as a derived signal in every output file: a shared fingerprint can reflect a shared family or workplace device, and the label was not collected with ring membership in mind. A cluster is called *credible* only if it has at least 5 transactions and a proxy rate at least $3\times$ the split's overall rate (4.57%). This two-part rule replaced an earlier size-only criterion that conflated large ordinary clusters with concentrated ones. Because it uses labels, it is an offline evaluation criterion; a label-free variant is used for live streams, where a new cluster's fraud rate is unknown.

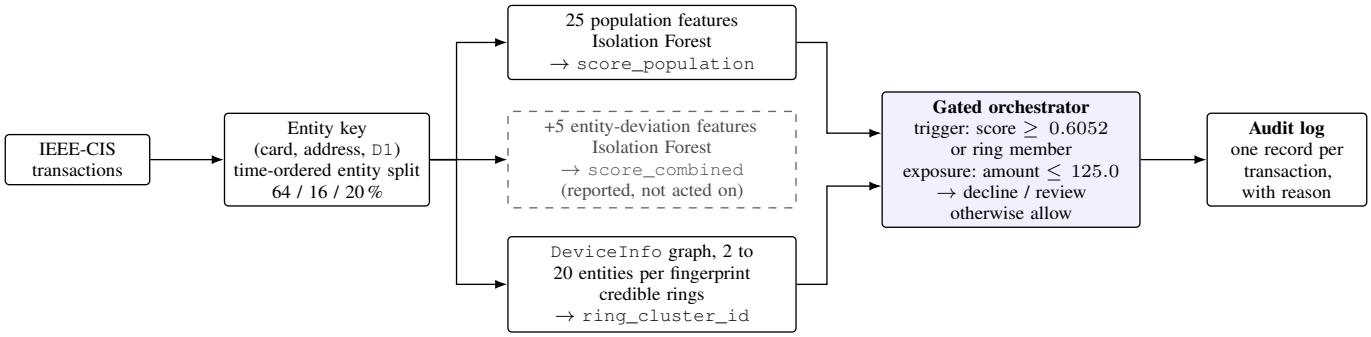


Fig. 1. Sentinel pipeline. The entity-deviation branch (dashed) is evaluated and logged but, per the result in Section IV-C, does not drive the gate. Thresholds shown are the values calibrated in this run.

F. Gated Response Orchestrator and Audit Log

The gate fires on either of two independent signals, never blended into one number: membership in one of the credible rings of Section IV-D, or a population score at or above a confidence threshold, calibrated as the 98th percentile of population scores on the validation split (0.6052 in this run). The confidence gate uses the population score rather than the combined score: since Section IV-C shows the entity-deviation feature does not help, the response layer does not act on it. A triggered transaction is then routed by *exposure*, taken to be its own amount (`TransactionAmt`), a direct proxy for how much is at stake. The autonomous-action threshold is calibrated, not hand-picked, as the 75th percentile of training-split amounts (125.0 in this run, in the dataset's native units; IEEE-CIS documents no currency for this field, so it is not a rupee or dollar figure). A triggered transaction at or below it is auto-declined; one above it goes to a human reviewer, on the reasoning that the more money is at stake, the less appropriate it is to act alone. A transaction that trips neither signal is allowed. Sentinel therefore acts autonomously only within a bounded exposure. Every outcome, including plain allows, is written to an audit log with a short, fixed-vocabulary reason naming the rule that fired.

IV. Experimental Results

A. Baseline Confusion-Matrix Performance

Table I reports the five baselines on the held-out test split with full confusion-matrix counts, so the cost of each method's false positives is visible directly. The entity-level rules recover higher recall than any population-level baseline (16.96% and 18.38% against a ceiling of 8.62%) but at a false-positive rate above 19%, three to six times that of the population-level methods. On entities with usable history, personalization finds more fraud, but at a cost that makes neither entity rule usable alone as an autonomous filter.

B. Detector Comparison by PR-AUC

Table II extends the comparison to the Isolation Forest. On the full test split the population-only configuration reaches a PR-AUC of 0.1656, 3.0 to 3.7 times every rule or statistical

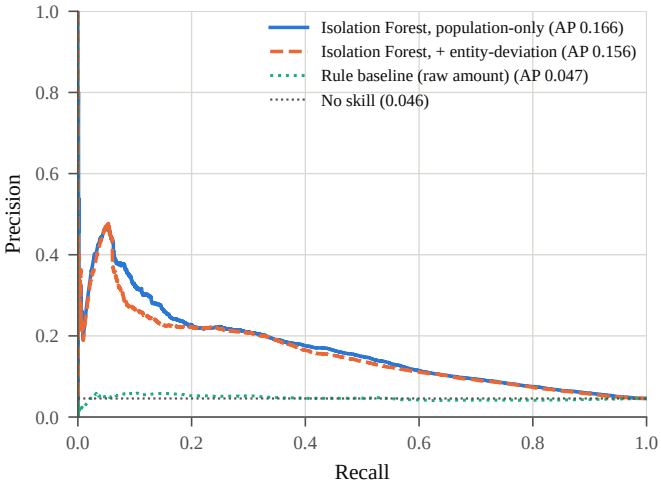


Fig. 2. Precision-recall curves on the full test split ($n = 85{,}004$), computed from the logged scores. AP is average precision.

baseline, none of which separates clearly from the random-prior floor of 0.0454 (Fig. 2). Adding entity-deviation features does not improve on this: it is lower on the full split (0.1557) and on has-history rows (0.0620 against 0.0639).

To attach uncertainty, test entities were resampled with replacement (1,000 entity-cluster bootstrap draws, fitted models held fixed). PR-AUC is 0.166 (95% CI 0.155 to 0.178) for population-only and 0.156 (0.146 to 0.168) for the combined configuration. The paired difference is 0.010 (0.0075 to 0.0124) and positive in all 1,000 draws, so the entity features reliably cost a little rather than merely adding nothing. Even the lower interval bound for the population-only detector, 0.155, is 2.8 times the best baseline's point estimate (0.0546). The bootstrap does not capture training-seed variance. One boundary is worth stating: on has-history rows alone the detector's advantage narrows sharply (0.0639 against 0.0555 for the best baseline), so most of its gain is earned on cold-start transactions.

C. History-Depth Diagnostic

The 85,004 test transactions come from 54,963 entities, 1.55 per entity on average. Among the 30,041 has-history rows the median entity has two prior transactions; 12,282 (40.9%)

TABLE I
RULE- AND STATISTICS-BASED BASELINES ON THE HELD-OUT TEST SPLIT

Baseline	Precision	Recall	FPR	TP	FP	FN	TN
Random prior	4.61%	3.06%	3.03%	119	2,460	3,766	78,659
Rule, population (amount $>$ p95)	5.62%	8.62%	6.93%	335	5,624	3,550	75,495
Statistical, population ($\lvert z\rvert > 2$)	5.04%	4.35%	3.93%	169	3,185	3,716	77,934
Rule, entity (new record high)*	2.98%	16.96%	21.54%	191	6,227	935	22,688
Statistical, entity ($\lvert z\rvert > 2$ vs. own history)**	4.03%	18.38%	19.31%	138	3,285	613	13,723

*Evaluated on the 30,041 has-history rows only (64.7% of the test split excluded as cold-start). **Evaluated on the 17,759 rows with two or more prior transactions only (79.1% excluded).

TABLE II
PR-AUC ON THE TEST SPLIT

Method	Full test	Has-history
Random prior	0.0454	0.0383
Rule, population (raw amount)	0.0467	0.0555
Statistical, population ($\lvert z\rvert$)	0.0546	0.0488
Rule, entity (ratio to prior max)	0.0467	0.0373
Statistical, entity ($\lvert z\rvert$ vs. own history)	0.0446	0.0382
Isolation Forest, population-only	**0.1656**	0.0639
Isolation Forest, + entity-deviation	0.1557	0.0620

have exactly one, too few to compute a meaningful deviation, and the remaining 17,759 (59.1%) have two or more. The statistical-entity baseline is itself restricted to this two-or-more population and still trails every population-level method, and the fingerprint construction and the split were each checked directly. The null result is therefore not an artifact of either: most entities here simply have too thin a history to personalize against. This is a property of the data's history depth, not a defect of the feature design, and it is reported rather than dropped.

D. Ring Detection at Full Test Scale

Linking over the full test split with fingerprints shared by 2 to 20 entities and clusters capped at 50 entities yields 516 clusters; none reached the cap. Together they span 3,371 transactions (4.0% of the split), so this layer touches a small fraction of traffic. Because clusters form without labels, their pooled fraud rate is a selection-free measure of linkage signal: 391 fraud-labeled transactions among 3,371, or 11.6%, 2.5 times the 4.57% base rate.

The credibility rule (at least 5 transactions and at least $3\times$ the base rate) keeps 70 of the 516 clusters (13.6%), spanning 760 transactions (Fig. 3). Seven reach 100% concentration, at up to 12 entities, and one 18-entity cluster reaches 77.8% (14 of 18 transactions). Two cautions apply. First, selection by label guarantees an elevated rate: the 41.6% pooled fraud rate of the 70 credible clusters (316 of 760) is selection-inflated and is not a detection precision. Second, a chance baseline is needed. If labels fell on transactions independently at the base rate, the same filter would pass 34 ± 5 of the 516 clusters (maximum 50 in 2,000 simulations). The observed 70 is a real excess, but about half of the credible set is what chance alone would supply. The independence null ignores the clustering of fraud within entities, which makes chance passes more common, so

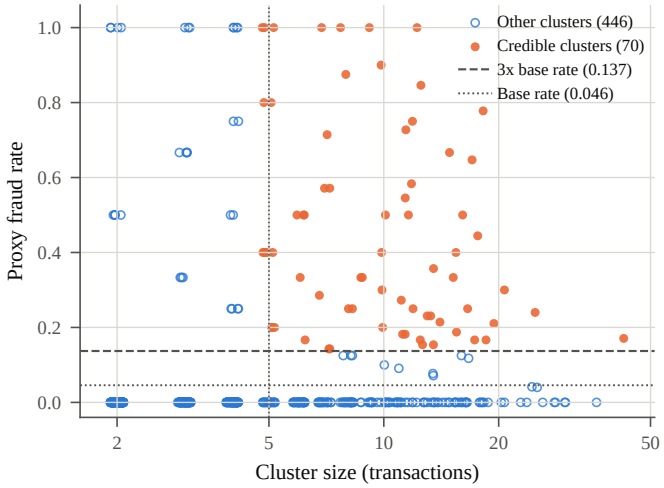


Fig. 3. Cluster size against proxy fraud rate for all 516 device-fingerprint clusters. The 70 credible clusters (orange) meet both parts of the credibility rule. Rates are derived proxies, not verified ground truth.

the excess is indicative rather than a significance test. The rule is one operating point: a $4\times$ multiple keeps 59 clusters, $2\times$ keeps 76, a transaction floor of 3 keeps 92, and a floor of 10 keeps 38, with the same qualitative picture of a small concentrated subset above a long ordinary tail.

E. Gated Response and Audit Completeness

The orchestrator was run over all 85,004 test transactions, the same split used throughout Section IV, with the 70 credible rings and the thresholds of Section III-F. Table III summarizes the outcome. 760 transactions (0.9%) were ring-linked and each was declined (656) or escalated (104). A further 1,499 transactions tripped only the score trigger (1,218 declined, 281 escalated); of the ring-linked, 705 would have passed the score gate alone. The remaining 82,745 (97.3%) were allowed.

Decision quality needs care because ring credibility uses labels. The ring-linked fraud rate of 41.6% is selection-inflated and should not be quoted as precision. The label-free number is the score trigger: 414 of its 1,499 transactions are fraud-labeled, 27.6%, six times the base rate. Across both triggers the gate catches 730 of 3,885 fraudulent transactions (18.8% recall) while touching 2.7% of traffic, and the allowed set carries 3.8% fraud. It is deliberately conservative, a high-precision audited filter rather than a high-recall one.

TABLE III
GATE OUTCOMES ON ALL 85,004 TEST TRANSACTIONS

Group	Txns	Decl.	Rev.	Fraud	Rate
Ring-linked†	760	656	104	316	41.6%
Score trigger only	1,499	1,218	281	414	27.6%
Allowed	82,745	–	–	3,155	3.8%
All	85,004	1,874	385	3,885	4.6%

†Credible rings are selected using test labels, so this rate is selection-inflated.

For audit completeness, the log holds exactly one record per test transaction (85,004 unique identifiers, none missing). Replaying the gate rules from each record's logged inputs (population score, ring identifier, amount) against the two calibrated thresholds reproduces all 85,004 decisions and reason strings with zero discrepancies, and a single pair of thresholds is consistent with every logged outcome. The log and the results store are written from the same decision object, so their agreement shows completeness; the replay is what verifies the logged decisions against the rules. Separately, a 123-transaction synthetic stream with an injected ring, built for the project's live console and never blended into any scored claim, exercises the same orchestrator end to end: 13 transactions are ring-linked, 110 allowed, 11 declined, and 2 escalated.

V. DISCUSSION AND LIMITATIONS

Two common shortcuts in fraud pipelines, personalizing a score without checking whether there is enough history to personalize against, and calling a device-linked cluster a ring without flagging the label as a proxy, both fail quietly unless checked directly. Checking the first produced a data-grounded explanation rather than an unexplained null; checking the second means every ring statistic carries its proxy caveat where it is used. The gate follows the same caution: because the ring label is a proxy, autonomy is bounded by exposure, so higher-stakes cases reach a human instead of resting on an unverified label.

Several limitations bound the claims. The entities and ring labels are proxies, and credibility is judged with test labels, so ring results show that shared fingerprints correlate with fraud, not a validated ring-detection accuracy. The ring layer covers only 4.0% of transactions. Decision quality is measured against the same dataset's label, and whether 27.6% precision at 18.8% recall is the right operating point is a policy question for domain stakeholders. The exposure threshold is one flat cutoff (the 75th percentile of training amounts) applied to every entity; whether it should scale per entity is an open design question. Finally, results come from one benchmark, the feature set omits the D, M, and V columns, and the bootstrap holds the fitted models fixed.

VI. CONCLUSION AND FUTURE WORK

Sentinel combines an entity- and population-aware anomaly scorer, a bounded-prevalence device-fingerprint ring detector, and a gated, audited response orchestrator, and holds all three to one standard: report the baseline, diagnose rather than hide a null result, and label every derived signal as derived. The Isolation Forest triples the best baseline's PR-AUC; entity-deviation features slightly reduce it, traced to thin per-entity history; fingerprint clusters carry 2.5 times the base fraud rate; and the gate's logged decisions replay exactly across the full test split. Future work includes validating ring concentration against independently labeled ring data, testing whether the entity-deviation null is specific to this benchmark by repeating it on data with deeper per-entity history, replacing the label-based credibility criterion with a label-free one throughout the evaluation, and choosing the gate's operating point with domain stakeholders.

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